

SECURE PASSWORD MANAGEMENT SYSTEM

PRG800 – Milestone 1

Submitted to:
Professor Miles McDonald

Submitted by:
Secure Vault Group

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PRG800NAA.06353.2264
SENECA POLYTECHNIC

May 11th, 2026

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1. Executive Summary

This milestone report presents the proposed **Secure Password Management System**, a cybersecurity-focused application designed to provide users with a secure and centralized platform for storing and managing passwords and sensitive information. The system addresses common security challenges faced by individuals who manage multiple online accounts and sensitive credentials.

Many users continue to rely on unsafe password practices such as reusing passwords, storing credentials in unsecured files, or creating weak passwords that expose them to cybersecurity threats and data breaches. The Secure Password Management System aims to reduce these risks by introducing secure authentication, encrypted credential storage, password generation, and password security monitoring features.

The proposed system will allow users to securely organize credentials while improving password management practices through secure software engineering principles such as encryption, hashing, authentication, and access control. Additional features such as password health analysis, secure file storage, two-factor authentication, and audit logging will further strengthen user security and system reliability.

This project combines cybersecurity concepts with project management practices, allowing the team to apply technical, organizational, and collaborative skills while developing a practical and real-world security solution.

2. Introduction

The purpose of this milestone report is to present the initial planning stage of the PRG800 group project. The report outlines the proposed project idea, identifies each group member and their responsibilities, defines the communication strategy, establishes the group contract, and presents the overall mission and vision for the system being developed.

The Secure Password Management System is intended to provide users with a reliable and secure platform for managing passwords and sensitive credentials. The project focuses on improving

digital security practices through secure authentication, encrypted storage, and password management tools that help users protect sensitive information from unauthorized access.

This project aligns strongly with cybersecurity and secure software development concepts by incorporating encryption, hashing, authentication, secure storage, and access management techniques. In addition, the project allows the team to apply project management principles, including planning, communication, collaboration, risk analysis, testing, and documentation.

3. Problem Statement

Many individuals struggle to securely manage multiple passwords and sensitive credentials across various online platforms. Common unsafe practices such as password reuse, weak password creation, storing passwords in unsecured documents, or relying on memory increase the likelihood of unauthorized access, credential theft, and data breaches.

Current password management habits often expose users to:

- Password fatigue
- Weak authentication practices
- Security vulnerabilities
- Unsafe storage habits
- Sensitive information exposure

Without a secure centralized solution, users face difficulty maintaining strong password hygiene and protecting confidential information. The proposed Secure Password Management System addresses these challenges by providing encrypted password storage, secure authentication, password generation, and additional security-focused features that improve overall digital safety and user convenience.

4. Personal Profile & Group Communications

Group Name: Secure Vault

Group Members / Individual Profile

NAME	EMAIL	SKILLS	ROLE
Felix Akuetteh Torto	fatorto@myseneca.ca	Networking, documentation, and coordination	Project Leader
Akshay Nedumparambil Unnikrishnan	anedumparambil-unnik@myseneca.ca	Networking, testing, research	Research Analyst/ documentation
Harmon Anthony Tuazon	hamtuazon@myseneca.ca	Web development, cloud development, and database administration	PM / Developer / Programmer
Jean Luc Sita Mbuya	sjlmbuya@myseneca.ca	DevOps, cloud computing, database design	Technical Lead / Database Designer
Kulsum Timol	katimol@myseneca.ca	Web development, programming, and database design	Developer / Programmer
Anju Babu	ababu35@myseneca.ca	Full-stack development, QA testing	Developer / QA Tester

Group Communications Plan

WhatsApp	Daily communication and quick updates
Weekly virtual meetings	Review project progress and discuss tasks
GitHub	Version control and collaborative development
OneDrive / Google Drive	Documentation and file sharing

Communication Expectations:

- Members are expected to communicate delays early.
- Respond to important messages within 24 hours.
- Project responsibilities and progress updates will be reviewed during group meetings.

5. Integrity and Professionalism

All group members agree to maintain academic integrity, professionalism, and ethical behavior throughout the project's duration. Each member is responsible for ensuring that submitted work is original and that all external references and resources are properly acknowledged.

The group understands that academic integrity applies to all members involved in the project collectively. Any academic misconduct or violation of Seneca Polytechnic policies may affect all members associated with the submission. Therefore, all members agree to contribute fairly, communicate professionally, and uphold ethical standards in all project activities.

6. Group Contract

Working Practices Contract

By signing below, we acknowledge having read the Seneca Academic Integrity Policy as well as the above conditions.

Felix Akuetteh Torto		168365229
Akshay Nedumparambil Unnikrishnan		190635235
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Kulsum Timol		112867247
Anju Babu		115640245

Group Name: Secure Vault

Responsibilities of the Project Leader Include:

- Assigning tasks to other group members, including oneself, in a fair and equitable manner
- Ensuring work is completed with accuracy, completeness, and timeliness
- Planning for task completion to ensure timelines are met
- Any other duties as deemed necessary for project completion

Group Commitments

All group members agree to:

- Contribute fairly to the project.
- Complete assigned tasks on time.
- Communicate respectfully and professionally.
- Support team members facing genuine difficulties
- Participate actively in meetings and project decisions.
- Follow academic integrity guidelines.
- Maintain collaborative teamwork throughout the project lifecycle.

7. Scenario Agreements

Scenario	Accepted Y/N + initial	We agree to do the following
Group member does not deliver component on time due to severe illness or extreme personal problem	Y (FAHJKA)	a) Group absorbs workload temporarily ____ b) Group seeks advice from professor ____ c) Group shifts target date if possible ____ d) Other:
Group member cannot deliver component on time due to lack of ability	Y (FAHJKA)	a) Group reassigns component ____ b) Group helps member ____ c) Group member must ask professor for reference material ____ d) Other:
Group member does not deliver component on time due to lack of effort	Y (FAHJKA)	a) Group absorbs workload ____ b) Other:

Scenario	Accepted Y/N + initial	We agree to do the following
Group member does not attend group meeting	Y (FAHJKA)	a) Group proceeds without him/her and will assign work to the absent member ____ b) Group doesn't proceed and records group member's absence ____ c) Other ____
A piece of production equipment fails such as a laptop	Y (FAHJKA)	a) Backup copies will be made and kept with the group leader ____ b) GitHub or similar source control will be used ____ c) Other:
An unforeseen constraint occurs after the deliverable has been allocated and scheduled (a surprise test or assignment)	Y (FAHJKA)	a) Group meets and reschedules deliverable ____ b) Group will cope with constraint ____ c) Other:
Group cannot achieve consensus leaving one member feeling "railroaded", "ignored", or "frustrated" with a decision which affects all parties	Y (FAHJKA)	a) Group agrees to abide by majority vote ____ b) Group flips coin ____ c) Other:
Group members do not share expectations for grade desired	Y (FAHJKA)	a) Group will elect one person as "standards-bearer" who has the right to ask that work be redone ____ b) Group votes on each submission's quality ____ c) Other:

Scenario	Accepted Y/N + initial	We agree to do the following
Group member behaves in an unprofessional manner by being rude or uncooperative	Y (FAHJKA)	a) Group attempts to resolve the issue by airing the problem at group meeting ____ b) Group requests meeting with professor to problem-solve ____ c) Group ignores behaviour ____ d) Group agrees to avoid use of all vocabulary inappropriate to the business setting ____
There is a dominant group member who is content to make all decisions on the group's behalf leaving some group members feeling like subordinates rather than equal members	Y (FAHJKA)	a) Group will actively solicit consensus on all decisions which affect project direction by asking for each member's decision and vote ____ b) Group will express subordination feelings and attempt to resolve issue ____ c) Other:
Group has a member who refuses to participate in decision making but complains to others that s/he wasn't consulted	Y (FAHJKA)	a) Group forces decision sharing by routinely voting on all issues ____ b) Group routinely checks with each other about perceived roles ____ c) Group discusses the matter at group meeting ____

8. Group Mission Statement

Our mission is to collaboratively design and develop a secure Password Management System that enables users to safely store, organize, generate, and manage passwords and sensitive information through encryption, authentication, and secure software engineering practices while maintaining professionalism, teamwork, and academic integrity.

9. Group System Vision

Our vision is to create a secure, reliable, and user-friendly password management platform that improves digital security practices for individual users and students. The system will provide encrypted credential storage, secure authentication, password generation, password health monitoring, sensitive file protection, and security-focused features that reduce cybersecurity risks and promote safer password management habits.

Key System Features

Secure user authentication	Verifies user identity before granting access
Password hashing and encryption	Protects stored credentials from unauthorized access
Password generator	Creates strong, random passwords for users
Password strength analysis	Evaluates and suggests improvements for weak passwords
Secure vault storage	Centralized encrypted storage for all credentials
Two-factor authentication (2FA)	Adds an extra layer of security
Password health dashboard	Displays the security status of stored passwords
Auto logout functionality	Automatically ends sessions after inactivity
Audit log tracking	Records access and changes for security monitoring

10. Overall Project Goal

The overall goal of this project is to improve user cybersecurity by developing a secure centralized platform for managing passwords and sensitive information. The system aims to reduce the risks associated with weak password practices, insecure storage methods, and unauthorized access while promoting safer digital habits and stronger authentication security for users.